

CHE 330 Week 2 Homework

Assigned 9/3/2025, Due 9/9/2025 11:59 PM

1. Define and indicate the formulas and notations for the Number, Weight, and z-Average Molecular Weights of a polymer

Number: Average weight determined by number of molecules

Weight: The average molecular weight calculated by weighing each polymer chain in proportion to its mass contribution. Larger chains contribute more to the average.

Z-average: A higher-order average molecular weight that emphasizes the contribution of the largest molecules even more.

$$\bar{M}_z = \frac{\sum n_i M_i^3}{\sum n_i M_i}$$

$$\bar{M}_n = \frac{\sum n_i M_i}{\sum n_i}$$

$$\bar{M}_w = \frac{\sum n_i M_i^2}{\sum n_i M_i}$$

✓ 3

- a. Polydispersity Index and Degree of Polymerization

$$PDI, PI = \frac{\bar{M}_w}{\bar{M}_n} \quad \text{measure of how broad molecular wt distribution of polymer is}$$

✓

$$(DP)_n = \frac{M}{M_0} \quad \text{the number of repeating units in the polymer}$$

✓

Chain

* molecular wt polymer over molecular wt of the monomer

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2. During a polymer synthesis experiment, your team members reported the following data for a fractionated polyester made from oxalic acid and 1,6-hexanediol. You have been assigned to compute the M_n , M_w , M_z , PDI and σ of the polyester formed.

Fractions	1	2	3	4	5	6	7	8	9
Mass (g)	1.082	0.904	0.801	0.310	0.041	1.043	1.281	1.171	0.241
$M \times 10^{-4}$	1.042	2.312	2.619	3.283	3.501	3.984	5.017	7.984	9.308

n_i : 1.038 2.391 2.306 0.099 0.0117 0.262 0.255 0.147 0.026

$$\sum_{i=1} n_i M_i = 6.876$$

$$\sum_{i=1} n_i M_i^2 = 28.67 \quad (\times 10^5)$$

$$\sum n_i M_i^3 = 159.865 \quad (\times 10)$$

$$\bar{M}_n = \frac{\sum n_i M_i}{\sum n_i} = \frac{6.876}{2.7187} = 3.031 \text{ g/mol}$$

X

$$\bar{M}_w = \frac{\sum n_i M_i^2}{\sum n_i M_i} = \frac{28.67}{6.876} = 4.1696 \text{ g/mol}$$

✓

159.865

$$M_z = \frac{\sum n_i M_i^3}{\sum n_i M_i^2} = \frac{\quad}{28.67} = 5.576 \text{ g/mol}$$

$$PDI = \frac{\overline{M}_w}{\overline{M}_n} = \frac{4.1696}{3.031} = 1.376$$

$$G = \overline{M}_n \left(\frac{\overline{M}_w}{\overline{M}_n} - 1 \right)^{1/2} = 1.859$$

3. A polymer mixture P has number-average and weight-average molecular weights of 178,000 and 334,603.93 respectively and was divided into fractions A and B by fractional precipitation.

- a. If A and B have number-average molecular weights of 128,000 and 225,000 respectively, what are the weight fractions of A and B obtained from the initial polymer P?

$$\frac{1}{178,000} = \frac{w_A}{128,000} + \frac{w_B}{225,000} \quad | \quad w_B = 1 - w_A$$

$$w_A = 0.348$$

$$w_B = 0.652$$

- b. If A and B have the same polydispersity, what is the polydispersity index?

$$\frac{\overline{M}_{w,P}}{\overline{M}_{n,P}} = K, \quad \overline{M}_{w,P} = w_{t,A} \cdot \overline{M}_{w,A} + w_{t,B} \cdot \overline{M}_{w,B}$$

$$= K (w_{t,A} \cdot \overline{M}_{n,A} + w_{t,B} \cdot \overline{M}_{n,B})$$

$$K = \frac{\overline{M}_{w,P}}{w_{t,A} \overline{M}_{n,A} + w_{t,B} \cdot \overline{M}_{n,B}} = 1.75$$

4. What would be Mw, Mn and polydispersity for a sample obtained by mixing 25 g of polystyrene (Mw = 75,000 g/mol; Mn = 65,670g/mol) with 30 g of another polystyrene (Mw = 88,000 g/mol; Mn = 45,000 g/mol)?

Problem 4

wt% 1 0.454545455

wt% 2 0.545454545

Mn (g/mol) 52513.0852

Mw (g/mol) 82090.90909

PDI 1.563246737

$$\overline{M}_{N,mix} = \frac{1}{\frac{w_A^*}{\overline{M}_{N,A}} + \frac{w_B^*}{\overline{M}_{N,B}}}$$

$$\overline{M}_{w,mix} = w_A^* \overline{M}_{w,A} + w_B^* \overline{M}_{w,B}$$



	Mw	Mn	grams	
sty 1	75,000	65670	25	
sty 2	88,000	45000	30	

5. Calculate the number-average and weight-average molecular weights of the mixture below, be sure to include the correct formulas for these calculations:

Polyme r	Mn (g/mol)	Mw (g/mol)	Weight in Mixture (g)
A	35000	67250	1342
B	48000	58190	3245
C	65548	72549	2458

total : 7045

n_i w_i
 0.0383 0.191
 0.0676 0.461
 0.0375 0.348
 0.1434

$$M_n = \frac{\sum n_i M_i}{\sum n_i} \rightarrow n_i = \frac{n_i M_i}{M_n} \quad \text{and} \quad w_i = \frac{n_i M_i}{\sum n_i M_i}$$

$$\eta_h = \frac{7045}{0.1434} = 49131.2 \text{ g/mol}$$

$$w_i \cdot w_i = \frac{w_i}{w_i} \quad w_i = n_i \cdot m_i$$

$$m_w = \sum v_i \cdot h \cdot M_{w,i} = 64804.7 \text{ g/mol}$$

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6. (Extra Credit)

$$\overline{M}_{w,mix} = w_A^* M_{w,A} + w_B^* M_{w,B}$$

For a Mixture of Polymers A and B, prove the following: